

Age-Dependent Normalization Functions for T Lymphocytes in Healthy Individuals

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Age-Dependent Normalization Functions for T Lymphocytes in Healthy Individuals

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Lymphocyte numbers naturally change through age. Normalization functions to account for this are sparse and mostly disregard measurements from children in which these changes are most prominent. In this study, we analyze cross-sectional numbers of mainly T lymphocytes (CD3⁺, CD3⁺CD4⁺, and CD3⁺CD8⁺) and their subpopulations (naïve and memory) from 673 healthy Dutch individuals ranging from infancy to adulthood (0–62 y). We fitted the data by a delayed exponential function and estimated parameters for each lymphocyte subset. Our modeling approach follows general laboratory measurement procedures in which absolute cell counts of T lymphocyte subsets are calculated from observed percentages within a reference population that is truly counted (typically the total lymphocyte count). Consequently, we obtain one set of parameter estimates per T cell subset representing both the trajectories of their counts and percentages. We allow for an initial time delay of half a year before the total lymphocyte counts per microliter of blood start to change exponentially, and we find that T lymphocyte trajectories tend to increase during the first half a year of life. Thus, our study provides functions describing the general trajectories of T lymphocyte counts and percentages of the Dutch population. These functions provide important references to study T lymphocyte dynamics in disease, and they allow one to quantify losses and gains in longitudinal data, such as the CD4⁺ T cell decline in HIV-infected children and/or the rate of T cell recovery after the onset of treatment. *The Journal of Immunology*, 2023, 210: 1882–1888.

Lymphocyte numbers experience dynamical changes over a lifetime, known as a hallmark of a maturing and subsequently aging immune system (1–5). These dynamics are more prominent in childhood than in adulthood, as a child's immune system is maturing and has not yet approached steady state. These natural changes over time have to be considered when determining the immunological health status of an individual.

To define an individual's immunological health status, cell counts and percentages of various lymphocyte subsets are measured and compared with reference values of healthy individuals (6–10). For this reason, lymphocyte data of healthy individuals have been presented in contiguous age categories, combining lymphocyte measurements from several months to several years into bins (11–16). These age intervals allow clinicians to classify the health status of an individual at a given age.

As these bins often cover a broad age range and the lymphocyte numbers themselves are variable, continuous and gradual changes in lymphocytes over time during a disease—such as the decline in CD4⁺ T cell numbers during HIV infection and/or their recovery after the onset of treatment—are difficult to quantify. Instead of binning the data into age categories, one can also define continuous normalization functions as healthy references. By providing a finer time scale, such functions allow a better quantification of lymphocyte dynamics.

To our knowledge, only few publications present estimates describing lymphocyte data of healthy individuals in a continuous manner

(1, 17, 18). The often cited paper from Huenneke et al. (17) describes the trajectories of several T lymphocyte subsets and their subpopulations based on 100 healthy German individuals by fitting an exponential function, decaying toward an asymptote. Unfortunately, this dataset only includes 14 children under the age of 1 y, which is the time frame during which one expects major dynamical changes. Moreover, their study neglects an initial increase within the first half a year of age as observed by longitudinal analyses and in large cohort studies (5, 19, 20). Additionally, Huenneke et al. (17) provide neither parameters for the total lymphocyte counts (TLC) nor for the CD4⁺ and CD8⁺ percentages. Percentages usually show less variability than the cell counts per microliter of blood (21), and they are easier and less expensive to obtain through flow cytometry analysis (12). Thus, percentages are more commonly reported than counts. As percentages cannot replace the information retrieved from count data, normalization functions for both counts and percentages are required.

To extend on this and to obtain a much more detailed description of the T lymphocyte dynamics during the first year of life, we fitted cross-sectional age-matched total lymphocyte and T lymphocyte measurements from 673 healthy Dutch individuals (ranging from 0 to 62 y), including 268 individuals under the age of 1 y. As a statistical improvement, our methods follow laboratory measurement procedures, where most cell counts are calculated from observed percentages within a truly counted reference population. We therefore fit percentage data, while receiving parameters describing the cell count

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Abbreviation used in this article: CD4ct, CD4 count; TLC, total lymphocyte count.

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data. Thus, to our knowledge, we provide new standard functions expressing both absolute and relative data for several T lymphocyte subsets and their subpopulations of the Dutch population.

Materials and Methods

Data

We accessed the age-matched raw data of two published Dutch cohorts: Dutch I (presented by circles in all figures) (12) and Dutch II (presented by triangles in all figures) (13). Both publications presented their lymphocyte data (T cells, B cells, and NK cells) in age categories. The data presented in Dutch I are typically used as standard reference intervals for the Netherlands and Europe (11, 17, 18). For Dutch I, we had measurements from 426 individuals including TLC as well as the counts and percentages of the CD3⁺, CD3⁺CD4⁺, and CD3⁺CD8⁺ subsets within the total lymphocyte gate. For Dutch II, we had measurements from 112 children and adolescents including TLC as well as the counts and percentages of the CD3⁺CD4⁺, CD3⁺CD8⁺, naive CD4⁺ (CD27⁺CD45RO⁻CD4⁺), memory CD4⁺ (CD45RO⁺CD4⁺), effector CD4⁺ (CD27⁻CD45RO⁻CD4⁺), naive CD8⁺ (CD27⁺CD45RO⁻CD8⁺), memory CD8⁺ (CD45RO⁺CD8⁺), and effector CD8⁺ (CD27⁻CD45RO⁻CD8⁺) subsets. To cover the naive, memory, and effector subpopulations until late adulthood, we accessed a third dataset (Dutch III; presented by squares in all figures) with an additional 135 individuals (22). For Dutch III, percentages and count data are given for the CD4⁺ and CD8⁺ T cell naive, memory, and effector subpopulations, but only count data are provided for CD4⁺ and CD8⁺ T cell subsets. Even though the cohorts were collected a decade apart in different laboratories, the quantifications of count data were based on similar methods (in Dutch I with a Coulter Counter model Z1, and in Dutch II/III with a Cell-Dyn Sapphire hematology analyzer), and we do not observe major differences by visual inspection of cross-sectional illustrations of the data (Figs. 1–3). Thus, we combined the three cohorts for our analysis and refer to them from now on as the Dutch cohorts.

Model

Absolute cell counts. The natural decline of lymphocyte counts over age has previously been described phenomenologically by an exponential decay toward an asymptote (17). To stay close to this previous analysis but still allow for a different initial phase, which has been observed in longitudinal data and large cohort studies (5, 19, 20), we extend the original model by defining two phases for the absolute cell counts per microliter of blood, $C(t)$, depending on time t in years (Fig. 1A). We decided to describe the initial phase (phase I) by a simple linear function that lasts until time T . This is succeeded by a second phase (phase II) describing the exponential toward an asymptote as reported by Huenecke et al. (17):

$$\begin{aligned} C(t) &= C_0 + \alpha t && \text{if } t \leq T, \\ C(t) &= (C_T - A) \times e^{-\lambda(t-T)} + A && \text{otherwise.} \end{aligned} \quad (1)$$

For any lymphocyte subset, C_0 is the initial cell count per microliter of blood at birth, which changes linearly until time T to a value C_T with a slope α (cells per microliter per year). From time T onwards the counts change exponentially from C_T at a rate λ (per year), and eventually approach an asymptote A (cells per microliter blood), which corresponds to the normal homeostatic level as seen in healthy adults. Depending on the initial value, C_0 , and the ultimate asymptote, A , our model 1 can describe either overall increasing or declining dynamics of different lymphocyte subsets. The exponential, λ , can be expressed as a half-life or doubling time, $t_{1/2} = (\ln 2/\lambda)$, giving an indication for the time required to approach the homeostatic level, A , from the level at time T , C_T . Because this is independent of the difference, $C_T - A$, in cell numbers, we also report the quantity $\lambda(C_T - A)$, which is the initial slope after the early initial phase in terms of cells per microliter per year. Model 1 can be fitted to the absolute cell count data of any lymphocyte subset.

From percentages to absolute cell counts. In immunology, one is typically interested in both the absolute cell counts and the relative fractions (typically presented as percentages) of lymphocytes subsets. Because percentages are easier and less expensive to measure by flow cytometry than counting cells directly, the data representing the percentages are more commonly gathered than the cell counts of a given subset. In experiments, one typically counts some reference population in a microliter of blood, such as total lymphocytes, and estimates the fractions of various subsets within that reference population by flow cytometry. The cell counts of the subsets are then calculated by multiplying these observed fractions with the observed cell

count of the reference population. We decided to fit the data according to this experimental procedure, and refrain from the conventional way of fitting model 1 directly to the, from now on referred, “calculated” cell counts.

Instead, we first fit model 1 to the cell counts of the reference population, $C_R(t)$ (in our case total lymphocytes). Keeping the estimated parameters of the reference population, we then fit the observed percentages, $P(t)$, of a particular subset, by reversing the calculation of the cell counts from the measurement procedure, that is,

$$P(t) = \frac{C(t)}{C_R(t)} \times 100, \quad (2)$$

while fitting the parameters of model 1 to describe the unobserved cell counts, $C(t)$, of the subset of interest (e.g., CD4⁺ T cells; see also Box 1). The advantages of this procedure are that 1) we stay close to the experimental measurement procedures, 2) we use the same parameters to describe the percentages and the cell counts of a particular subset, and 3) we can describe complicated trajectories of percentages that are decreasing and subsequently increasing over time with a very simple model. As a cross-check, we will show that fitting the calculated cell count data directly with model 1 provides similar estimates (see Supplemental Fig. S2, Supplemental Table II).

Algorithm and confidence intervals. To fit the models to the data, we use the Grind wrapper that is based on the FME package in R (version 4.0.2) (23, 24). Parameter estimates are obtained based on the Levenberg–Marquardt algorithm by minimizing the sum of squared residuals. We fit the logarithmically transformed data to account for the measurement variance of several magnitudes over age (particularly in the count data) and to receive normally distributed residuals. Parameter estimates are constrained to be positive and are tested on being significantly different from 0 by the F test at a significance level of 0.05 (as provided by the algorithm in the FME package). We additionally apply a bootstrapping method to obtain 95% confidence intervals for each parameter estimate by resampling the data 1000 times. The 95% confidence ranges for the parameters are displayed in Table I and Supplemental Table II.

Results

No direct decline in lymphocyte counts after birth

We modeled lymphocyte dynamics from birth to 60 y of age with a simple delayed exponential function (Eq. 1, Fig. 1A). We accessed published reference lymphocyte data of two Dutch cohorts. Together they included measurements from 538 healthy individuals covering the full age range. Measurements are enriched for individuals under the age of 1 y ($n = 253$), which allowed us to focus on T lymphocyte dynamics in the very early phase of life, when changes are largest. Longitudinal data and large cohort studies (5, 19, 20) report an initial increase in lymphocyte counts during the first half a year of age (rather than the typically assumed exponential decline from birth onward [17]). Visual inspection of our data (see Figs. 1B, 2, 3) and linear regression analyses, resulting in positive slope estimates for the data covering the first year of life (see Supplemental Table I), support these observations. Consequently, we have extended the conventional exponential model (17) with an initial phase allowing for an early increase (Eq. 1, Fig. 1A).

Fitting TLC as a reference population

An attempt to freely fit all five parameters of the model (Eq. 1) to the cross-sectional TLC resulted in high correlations of the parameters describing the first phase (C_0 , α , and T). Thus, these three parameters are not identifiable from our data. We therefore decided to fix T to the in-literature-reported minimum value of $T = 0.5$ y (5, 19, 20), which is identical with the freely estimated value for T and additionally reduces the number of free parameters in the model (Eq. 1) from five to four. The best fit of the TLC data results in a linearly increasing phase over the first half a year from $C_0 = 5406$ cells/ μ l to $C_T = 5988$ cells/ μ l, before the TLC starts to decline exponentially at a rate of $\lambda = 0.221$ per year toward an

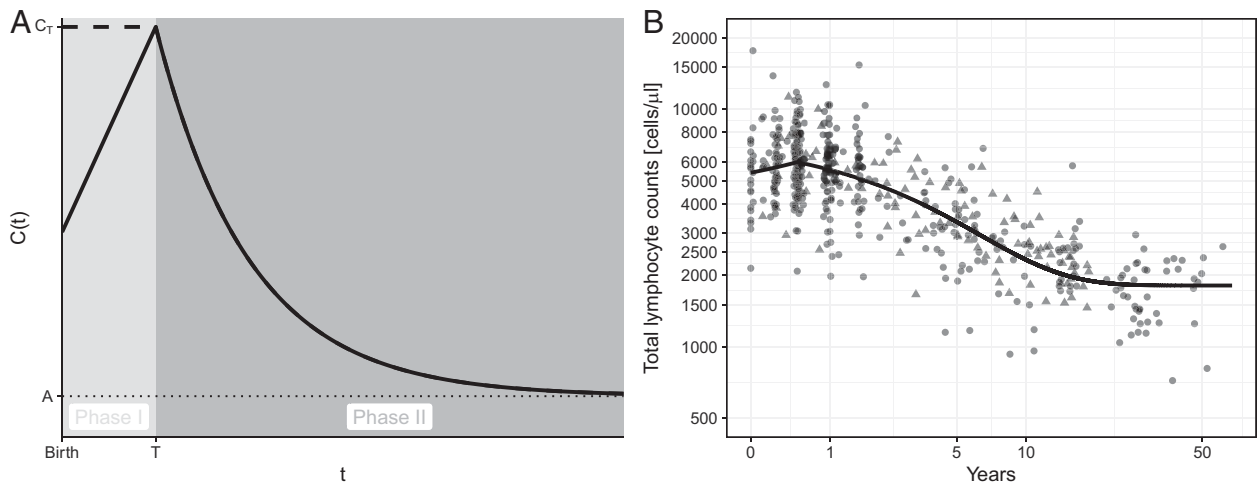


FIGURE 1. Lymphocyte trajectory follows a delayed exponential decay toward an asymptote. **(A)** Schematic illustration of the mathematical model (Eq. 1) with a flat (dashed line) and an increasing (solid line) initial linear phase (I) lasting until time T , followed by an exponential declining phase (II) from the value at $T(C_T)$ toward an asymptote A (dotted line) on a linear scale. **(B)** Measured total lymphocyte cell counts/ μl blood over age ($\bullet$, Dutch I; $\blacktriangle$, Dutch II) are shown along with the best fit of model 1 to the data (black line) on a logarithmic scale. The parameters used to receive the fits are listed in Table I.

asymptotic level of $A = 1810$ cells/ μl (Fig. 1B, Table I). Identifying the slope α of the initial phase remains difficult (see Table I), because α correlates with the initial value C_0 , and α is not significantly different from 0. This is due to the wide spread of the cross-sectional data suggesting interindividual variabilities. Given the good evidence for an initial increase in previous studies (5, 19, 20), we continued with the non-zero $\alpha = 1165$ cells/ $\mu\text{l}/\text{y}$, as we aimed to provide a TLC reference function for the Dutch population including the additional knowledge of an early increase. The parameters of the best fit implemented into model 1 result in a smooth trajectory describing the overall dynamics of the TLC data reasonably well (Fig. 1B, Table I). This smooth TLC trajectory served as the reference to obtain the following estimations for the trajectories of the CD3^+ , CD4^+ , and CD8^+ T cell subsets via model 2.

Box 1: Example calculations

Model 1 provides normalization functions for count data with the parameters presented in Table I. For example, for the reference population TLC and CD4 count (CD4ct) in cells per micro-liter:

$$\text{TLC}(t) = \begin{cases} 5405.5 + 1165.1 \times t & \text{if } t \leq 0.5 \text{ y,} \\ (5988.1 - 1810.1) \times e^{-0.221(t-0.5)} + 1810.1 & \text{otherwise;} \end{cases}$$

$$\text{CD4ct}(t) = \begin{cases} 2424.7 + 490.1 \times t & \text{if } t \leq 0.5 \text{ y,} \\ (2670.0 - 788.9) \times e^{-0.380(t-0.5)} + 788.9 & \text{otherwise.} \end{cases}$$

Model 2 provides normalization functions for the percentages. For example, for CD4%:

$$\text{CD4\%}(t) = \frac{\text{CD4ct}(t)}{\text{TLC}(t)} \times 100\%$$

$$= \begin{cases} \frac{2424.7 + 490.1 \times t}{5405.5 + 1165.1 \times t} \times 100\% & \text{if } t \leq 0.5 \text{ y,} \\ \frac{(2670.0 - 788.9) \times e^{-0.380(t-0.5)} + 788.9}{(5988.1 - 1810.1) \times e^{-0.221(t-0.5)} + 1810.1} \times 100\% & \text{otherwise.} \end{cases}$$

We first fit the function $\text{TLC}(t)$ to the TLC data, and subsequently fit the function $\text{CD4\%}(t)$ to the percentage data by estimating the four free parameters of the $\text{CD4ct}(t)$ function.

Fitting nonmonotonic percentages to predict cell counts

In contrast to the TLC, absolute counts of lymphocyte subsets, such as CD3^+ , CD4^+ , CD8^+ T cells, are typically calculated from the observed fraction within the lymphocyte gate; they are not directly counted. Although the calculated count data obey the trajectories described by model 1, the observed percentages of several lymphocyte subsets change in a more complicated manner over time, with an increasing and a decreasing phase (Fig. 2, left panels), which is more challenging to model. By following typical experimental procedures, and using model 2 with the TLC as a reference, we can describe these nonmonotonic trajectories with a rather simple function (Fig. 2, left panels). Thus, each trajectory can be estimated by four additional free parameters (C_0 , α , λ , and A), predicting the corresponding T cell counts by inserting them into model 1 (see Box 1).

Taking the CD4% as an example (Fig. 2B), one can recognize an almost flat phase during the first half a year at a value of 45%, before the CD4% drops by $\sim 10\%$ points to reach a nadir around the age of 5 y. Afterwards, the CD4% increases again to approach a plateau—at a level similar to the one observed at birth—in adulthood. Thus, a “wavy” CD4% trajectory is described, which confirms a previous observation (4). The CD3% and CD8% both expand from birth onward, and they also approach a homeostatic level around the age of 30 (Fig. 2A, 2C). Hence, by staying close to experimental procedures we model trajectories of the percentages that have more complicated dynamics than were previously appreciated.

Although our predictions for the calculated count data are based on fitting the observed percentages, they are in good agreement with the calculated CD3^+ , CD4^+ , and CD8^+ T cell counts (Fig. 2, right panels). Additionally, the trajectories predicted by our procedure of estimating observed data only are in excellent agreement with the trajectories that are obtained by estimating the calculated count directly with model 1 (Supplemental Fig. 1). Thus, the normalization functions obtained with either method (Table I, Supplemental Table II) are very similar.

Fitting complementary T cell subpopulations

Lastly, we provide estimates for the naive, effector, and memory subpopulations of CD4^+ and CD8^+ T cells (Fig. 3, Table I). Those subpopulations are usually reported as fractions of either CD4^+ or

Table I. Parameter estimates and their 95% confidence intervals to obtain fits for count data via model 1

Subset		C_0 (cells/ μ L)	α (cells/ μ L/y)	λ (1/y)	A (cells/ μ L)	$(C_T - A) \lambda$ (cells/ μ L/y)	$t_{1/2}$ (y)
Total lymphocyte count	Estimate	5405.5	1165.1	0.221	1810.1	923.33	3.14
	95% CI	[4794.0–5954.5]	[0.0–2785.9]	[0.174–0.287]	[1625.4–1986.5]	—	—
CD3 ⁺ T cell count	Estimate	3439.9	796.1	0.238	1287.0	607.13	2.91
	95% CI	[3221.1–3654.2]	[299.2–1326.3]	[0.218–0.262]	[1242.4–1330.0]	—	—
CD4 ⁺ T cell count	Estimate	2424.7	490.1	0.380	788.9	714.72	1.82
	95% CI	[2202.8–2622.8]	[0.0–987.6]	[0.341–0.427]	[755.5–821.5]	—	—
CD8 ⁺ T cell count	Estimate	963.7	211.8	0.156	386.5	106.56	4.44
	95% CI	[866.0–1057.6]	[0.0–440.5]	[0.133–0.179]	[350.5–418.6]	—	—
Naive CD4 ⁺ T cell count	Estimate	1733.1	0.0	0.162	428.1	211.41	4.28
	95% CI	[1447.6–2122.1]	[0.0–0.0]	[0.104–0.299]	[340.5–548.4]	—	—
Memory CD4 ⁺ T cell count	Estimate	140.4	167.2	0.600	332.1	–64.86	1.16
	95% CI	[49.2–228.3]	[0.0–454.4]	[0.004–10.476]	[309.9–999.9] ^a	—	—
Effector CD4 ⁺ T cell count	Estimate	6.8	0.0	0.000	6.8	0.00	—
	95% CI	[5.8–8.0]	—	—	[5.8–8.0]	—	—
Naive CD8 ⁺ T cell count	Estimate	607.5	208.9	0.079	169.6	42.85	8.77
	95% CI	[311.3–762.3]	[0.0–877.5]	[0.049–0.114]	[92.0–223.3]	—	—
Memory CD8 ⁺ T cell count	Estimate	22.2	57.3	0.468	122.4	–33.49	1.48
	95% CI	[0.0–55.3]	[0.0–156.9]	[0.037–1.073]	[109.8–175.2]	—	—
Effector CD8 ⁺ T cell count	Estimate	8.4	0.0	0.488	35.3	–13.13	1.42
	95% CI	[4.8–13.7]	[0.0–0.0]	[0.164–1.010]	[29.2–45.2]	—	—

The 95% confidence intervals for each parameter are received from bootstrapping the data 1000 times. The delay T was fixed to 0.5 y. The dash (—) shows that no CI or value was received through bootstrapping or calculation.
^aParameter was constrained to an upper limit of 1000 cells/ μ L.
CI, confidence interval.

CD8⁺ T cells and sum up to almost 100%. Unfortunately, these reference populations are typically not truly counted, which precluded us from using model 2. We therefore fitted the calculated count data using model 1 (Fig. 3, Supplemental Fig. 2, left panels, Table I). The three trajectories (naive, memory, and effector subpopulations) are then considered to sum up to 100%, from which we calculated trajectories for the percentages of the subpopulations according to model 2 (Fig. 3, Supplemental Fig. 2, right panels).

Our results (in black) fit the general trajectories of the data reasonably well (in gray). At birth the CD4⁺ and CD8⁺ T cell pools are largely composed of naive T cells (80–90%; Fig. 3). Naive T cell counts decline ~4-fold over 50 y, that is, from 1733 to 438 cells/ μ L for CD4⁺ and from 608 to 170 cells/ μ L for CD8⁺ T cells (Fig. 3, Table I). The memory compartments show opposite dynamics and increase over time toward an asymptote (Fig. 3). Overall, our trajectories are in good agreement with those reported by Huenecke et al. (17) (in gray). Note that Huenecke et al. (17) describe the memory CD4⁺ and naive CD8⁺ T cell counts by a linear model, whereas we also consider the effect of homeostasis in the subpopulations.

Discussion

We present, to our knowledge, a new mathematical approach to describe the natural dynamics in lymphocyte numbers continuously. By providing parameters for various T lymphocyte subsets (i.e., CD3⁺, CD4⁺, CD8⁺ T cells) and their subpopulations (naive and memory), reference functions for both their absolute counts and percentages can be formulated. Based on an extensive dataset, including age-matched cross-sectional lymphocyte measurements of 673 healthy Dutch individuals (covering particularly the dynamic first year of life), we are able to describe all trajectories phenomenologically with a biphasic model. We allowed for an initial phase of half a year in which lymphocyte subsets can increase, followed by a second phase that describes an exponential that approaches an asymptote corresponding to the homeostatic adult level. Except for memory T cells, this homeostatic level is below the values at birth. Our approach of fitting percentages to the ratio of two simple exponential models allowed us to simultaneously describe the counts and percentages of all populations of interest, delivering nonmonotonic trajectories of the

CD4⁺ and CD8⁺ T cell percentages. We provide a consistent set of parameters describing both percentages and their dependent counts.

We extended the model of a similar previous publication (17) with an initial phase to relax the previous assumption that the TLC declines from birth onward. Longitudinal data, as well as large cross-sectional studies, demonstrated that cell numbers per microliter of blood increase during the first half a year of life (5, 19, 20). We checked for such an early increase in our own data by linear regression within selected early time windows (Supplemental Table I). As these slopes were positive, we allowed for a first phase that lasts for the minimum reported value of 0.5 y. The parameters of the first phase have been difficult to identify, as the initial value, C_0 , and the slope parameter, α , are correlated. We are limited by the fact that the data are cross-sectional, and the data are variable between individuals. Accepting the strong external evidence for an initial increase, we decided to also accept the slopes estimated by the α parameter, even when these were not significantly different from 0.

We refrained from using the exponential initial phase proposed by Wade and Ades (1) and Payne et al. (18), because our linear model allowed us to easily define a fixed endpoint of the initial phase, and even this simple linear slope during a fixed first phase was difficult to identify. Thus, the extension of previous models with a fixed initial linear phase kept the model simple, while nevertheless allowing it to capture the different initial dynamics. We think the introduction of an initial phase is important when normalizing data from very young children. An example for using our reference functions in very young children is provided by Schröter et al. (25).

To our knowledge, our method is novel in the way that it stays close to experimental measurement procedures and only fits truly measured data. In most cases, these are percentages that are received via flow cytometry analysis. The conventional procedure to obtain cell count data is to multiply these percentages with the counts of a reference population (such as a TLC). Although count data tend to be variable, the calculated counts are even more variable because they combine the variation in the observed counts and the observed percentages. As a cross-check, we also fitted our model 1 to conventionally calculated count data (see Supplemental Fig. 1, Supplemental Table II), and indeed obtain a wider 95% confidence range of the estimated parameters (compare Table I with Supplemental Table II).

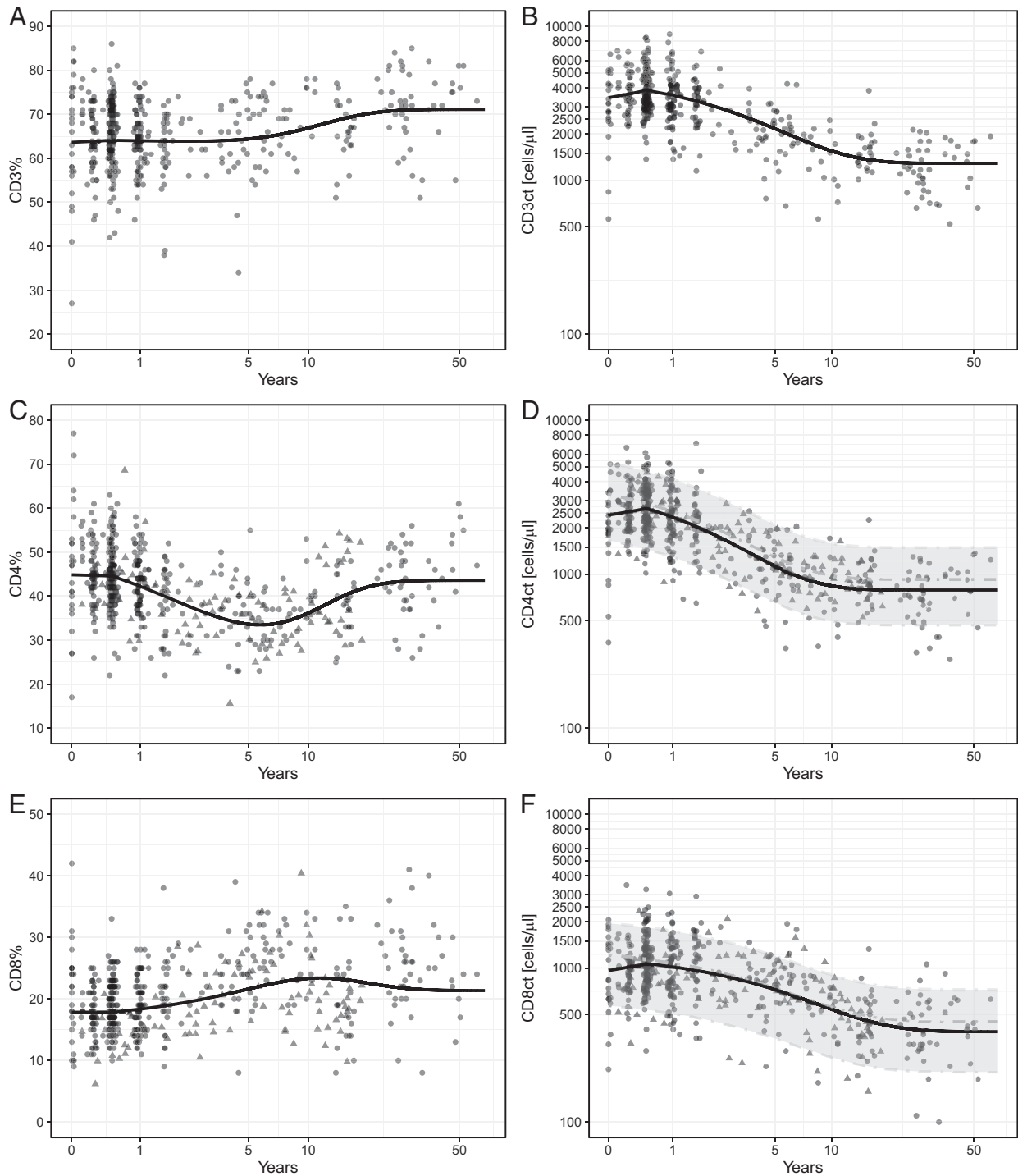


FIGURE 2. Trajectories of the T lymphocyte subsets CD3⁺, CD4⁺, and CD8⁺. (A and B) CD3⁺ cells (C and D) CD4⁺ cells. (E and F) CD8⁺ cells. (A), (C), and (E) illustrate the measured percentages; (B), (D), and (F) illustrate the calculated absolute counts. ●, Data from Dutch I; ▲, data from Dutch II. In (A), (C), and (E), the black solid lines represent the best fit of model 2. In (B), (D), and (F), the black solid lines represent the underlying trajectories for the calculated absolute count according to model 1. The parameters used to receive the fits are listed in Table I. When available, the estimates from Huenecke et al. (17) are presented in gray with their 95% confidence range.

Reassuringly, the average parameter estimates are in good agreement with each other, and as a consequence both methods provide very similar normalization functions.

Because we are aiming for normalization functions describing a population average, the main statistical question is what the best estimate is for the unobserved cell counts. The conventional calculated count has the advantage of using the count and percentage data from the same individual. This could be important if the counts and the

percentages are dependent variables. Because the measurement of counts is difficult, and count data show more interindividual variability than percentages, we decided to only model what is really measured. Fortunately, both fitting procedures provide very similar normalization functions. As techniques to count cell numbers are improving, one could in the future fit all count data directly via model 1. Normalization functions for the percentages would then be obtained by combining the model 1 parameters of the corresponding

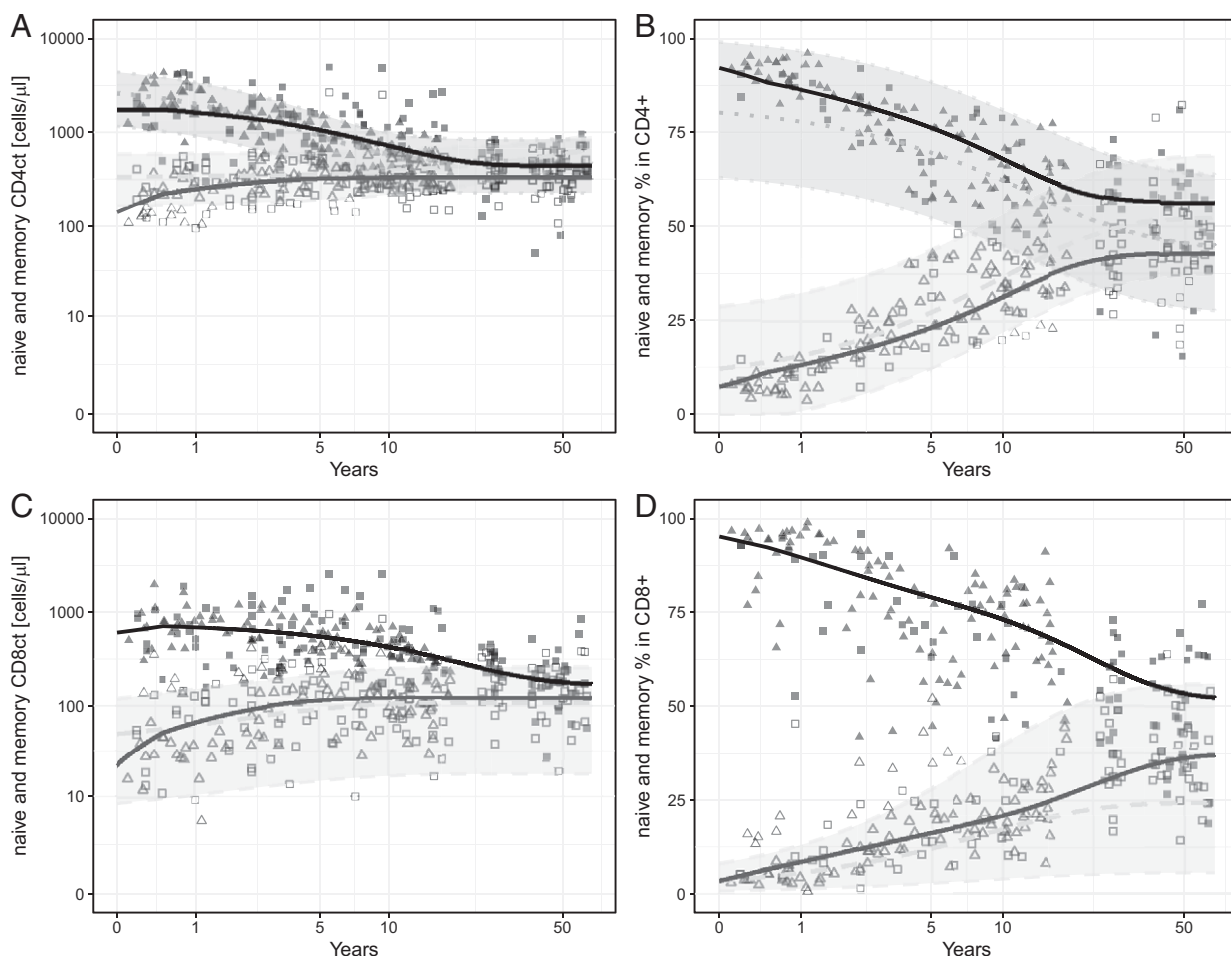


FIGURE 3. Trajectories of the naive and memory subpopulations of $CD4^+$ and $CD8^+$ cells. **(A and B)** $CD4^+$ cells. **(C and D)** $CD8^+$ cells. **(A)** and **(C)** give the absolute counts; **(B)** and **(D)** give the percentages. $\blacktriangle$ and $\triangle$, Data from Dutch II; $\blacksquare$ and $\square$, data from Dutch III. The naive subpopulation is represented by filled symbols, and the memory subpopulations are represented by open symbols. In **(A)** and **(C)**, the black solid lines represent the best fit of model 1 and the parameters reported in Table I. In **(B)** and **(D)**, the black solid lines represent the underlying trajectories for the percentages according to model 2 where the sum of naive, memory, and effector cell counts adds up to 100%. The darker shading refers to the naive subpopulation, the lighter shading refers to the memory subpopulations. The gray dotted (naive) and dashed (memory) lines represent the estimates from Huenecke et al. (17) and their 95% confidence range (gray area). We noticed that Huenecke et al. (17) described the percentages of the naive $CD8^+$ subpopulation by an increasing exponential function, whereas our data (and fit) follow a decline. We excluded this prediction from Huenecke et al. (17) in **(C)** and **(D)** (i.e., the missing gray range for naive $CD8^+$ counts), because we assume this to be a mistake (since the percentage of naive T cells should decline to general immunological knowledge). The parameters used to receive the fits are listed in Table I.

subpopulations using model 2, as we have done in gaining the trajectories for the percentages of the subpopulations. One could even think of fitting counts and percentages simultaneously to obtain the best population-based trajectories for counts and percentages.

We provide age-dependent continuous reference functions for lymphocyte data of both counts and percentages in the general Dutch population. Good agreement with a German reference (17) suggests that they might also serve as European references. We recommend to use these references for the Western/European population only, and do not recommend to use them for other populations. We expect major population differences in the lymphocyte trajectories due to environmental differences, for example, a more diverse and pathogen-rich environment in Africa compared with Europe (18). This would be important to prove, and it is therefore important to provide population-specific reference values also for other parts of the world. We are planning to repeat the analysis based on healthy data from South Africa (14).

A clear limitation of our work is the use of cross-sectional data, which only allows us to determine the dynamics of a population average. The variation around this average suggests that individuals

differ considerably in their T cell dynamics, and it would be of interest to quantify this variation between healthy individuals in terms of the parameters of our functions. This is particularly true for our estimates of the initial slope α that was difficult to estimate on the basis of the cross-sectional data, and we therefore fixed $T = 0.5$ based on previous studies (5, 19, 20). Estimating individual parameters would require longitudinal data, which are difficult to obtain, particularly in healthy children.

In conclusion, we provide age-dependent continuous reference functions for lymphocyte data of healthy individuals that can be used for normalization purposes. Our analysis does not replace the representation of lymphocyte data in contiguous age intervals, which is necessary to clinically determine the immunological health status of an individual, but rather provides an additional continuous presentation of the data. Age normalization gains importance when quantitatively assessing changes in diseases associated with the immune system. Having a standard normalization function at hand also simplifies the comparison of figures across different age categories by correcting for the confounding age effect, as done in Schröter et al. (25), where we quantified $CD4^+$ T cell recovery in antiretroviral therapy-treated HIV-infected children based on normalized $CD4$ counts.

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Disclosures

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